



GeoSight: An Automated Land Cover Classification and Change Detection Application for Rundu, Namibia (1994–2024)

**Student Name: Justine Shimpiku
Student Number: 223065560
Module: GIS Programming**

1. Executive Abstract

Monitoring land cover change is essential for environmental management, urban planning, and sustainable development. Traditional land cover classification workflows often require extensive fieldwork, manual data processing, and specialised GIS expertise. This project presents **GeoSight**, a desktop-based geospatial application developed to automate land cover classification and change detection using multi-temporal Landsat satellite imagery.

The application integrates image preprocessing, spectral index generation, machine-learning classification, change detection, and statistical reporting into a single workflow. Users provide multi-temporal satellite images, and the system automatically generates classified land cover maps, change maps, comparison maps, area statistics, and visual reports.

The application was tested using Landsat imagery covering the Rundu area of the Kavango East Region, Namibia, for the years 1994, 2004, 2014, and 2024. Results indicate substantial urban expansion over the study period, accompanied by a reduction in surrounding vegetation cover. GeoSight demonstrates how GIS programming can simplify complex remote sensing tasks by providing a fast, repeatable, and user-friendly solution for land cover monitoring.

2. Study / Project Background

Land cover information provides valuable insights into environmental conditions and human-induced landscape changes. Accurate mapping of built-up areas, vegetation, water bodies, and bare land supports decision-making in urban planning, natural resource management, and environmental monitoring.

Rundu, located in the Kavango East Region of Namibia, has experienced significant population growth and infrastructure development over recent decades. Expansion of residential areas, commercial activities, and transportation infrastructure has altered the surrounding landscape, making the town an ideal location for studying long-term land cover dynamics.

The motivation for this project originated from a previous remote sensing assignment that required extensive fieldwork to collect ground truth data using GPS devices. Although field validation remains important, the process required substantial time, travel, and financial resources. This experience highlighted the need for a more efficient approach to land cover analysis.

GeoSight was therefore developed to automate many of the processes traditionally performed manually, including classification, area estimation, change detection, and report generation. The project demonstrates how GIS programming techniques can improve efficiency while maintaining analytical reliability.

3. Objectives and Justification of the Project

Objectives

The objectives of this project were:

1. To develop a desktop application capable of automatically classifying land cover from multi-temporal satellite imagery.
2. To implement change detection functionality for identifying land cover transitions between different years.
3. To generate automated outputs including classification maps, change maps, area statistics, charts, and reports.
4. To apply the developed application to the Rundu study area using Landsat imagery from multiple years.
5. To evaluate urban expansion and other land cover changes occurring within the study area.

Justification

Traditional land cover classification workflows require significant user intervention and often involve multiple software platforms. Image preparation, classification, accuracy assessment, and change detection are usually performed separately, increasing both processing time and the possibility of user error.

GeoSight addresses these challenges by integrating all major analysis stages into a single application. The system improves efficiency, reproducibility, and accessibility while reducing the technical knowledge required to perform remote sensing analyses. The resulting workflow is particularly useful for students, researchers, planners, and environmental managers who require rapid land cover assessments.

4. Materials and Methods

Data Acquisition

Multi-temporal Landsat Collection 2 Surface Reflectance imagery was obtained using Google Earth Engine. Images representing the years 1994, 2004, 2014, and 2024 were selected and exported after cloud filtering and image compositing. The imagery was subsequently clipped to the Rundu study area using ArcGIS Pro.

The datasets consisted of six spectral reflectance bands together with the Normalised Difference Vegetation Index (NDVI) and the Normalised Difference Built-up Index (NDBI).

Application Development

GeoSight was developed in Python and packaged as a standalone desktop application. The application combines image preprocessing, classification, change detection, statistical analysis, and report generation within a graphical user interface.

The system automatically calculates additional spectral indices, prepares image inputs, trains a machine-learning classifier, performs land cover classification, evaluates classification accuracy, and generates visual outputs.

Classification Procedure

Land cover classification was performed using a Random Forest classifier. The model utilised spectral bands and vegetation, built-up, soil, and water indices to distinguish land cover categories.

Four land cover classes were identified:

- Vegetation
- Built-up Area
- Water
- Bare Land

Following classification, confidence-based refinement techniques were applied to improve classification consistency. Temporal comparison between years was then used to identify land cover transitions and quantify changes across the study period.

5. Description of Results / Model Application

GeoSight successfully processed all imagery and generated the required outputs automatically. These outputs included classified land cover maps, comparison maps, change detection maps, confidence maps, statistical summaries, and accuracy reports.

The classified maps revealed clear spatial patterns of land cover distribution within the study area.

Comparison of classifications from different years showed progressive urban expansion around the town centre and surrounding settlements. Vegetation remained the dominant land cover type throughout the study period; however, portions of vegetated land were converted to built-up areas as development increased.

Area statistics generated by the application indicated that built-up land increased steadily over time, while vegetation experienced a corresponding decline. The most significant transition identified by the change detection analysis was vegetation-to-built-up conversion, reflecting urban growth within the study area.

The application also generated exportable CSV tables and accuracy assessment reports, providing additional information for further analysis and validation. Overall, the outputs demonstrate the effectiveness of GeoSight in automating land cover mapping and change detection while maintaining a consistent and repeatable analytical workflow.

6. Concluding Remarks

This project successfully developed GeoSight, a desktop application for automated land cover classification and change detection using satellite imagery. The application integrates preprocessing, machine-learning classification, change detection, statistical analysis, and report generation within a single workflow.

Application of the tool to the Rundu study area revealed substantial urban expansion between 1994 and 2024, accompanied by reductions in surrounding vegetation cover. The generated maps, charts, and statistical outputs demonstrate the capability of the system to support environmental monitoring and urban growth assessment.

By reducing the need for extensive manual processing and field-dependent workflows, GeoSight provides a practical, efficient, and reproducible solution for land cover analysis. Future developments may include integration with higher-resolution imagery, real-time data processing capabilities, and web-based deployment to increase accessibility.

7. References

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